Ethiopian Association of Anesthetists

Clinical Audit Continuous Professional Development Training Module

Participant Manual

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Addis Ababa, Ethiopia

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Table of contents

Contents

Ethiopian Association of Anesthetists I

Clinical Audit Continuous Professional Development Training Module I

Participant Manual I

Table of contents 3

Acknowledgement ..... **Error! Bookmark not defined.**

List of tables and figures 6

List of Tables 6

List of figures 6

Forward 7

Approval statement 8

Acknowledgment ..... **Error! Bookmark not defined.**

Abbreviations and acronyms 9

Introduction to the manual 10

Scope of the manual 11

Core competency of the training 11

Rationale of clinical audit training (existing gaps and why addressing the gaps is important) 12

Module syllabus 13

Chapter 1: principles of clinical audit 17

Chapter description 17

Primary objective: 17

Enabling objectives: 17

Chapter outline 17

1.1 Introduction to clinical audit 17

Types of Clinical Audit 20

Difference between Clinical Audit, Research and Service Evaluation 23

1.2 Audit cycle 25

Chapter two: Clinical audit proforma 29

Chapter description: 29

Primary objective: 29

Enabling objectives: 29

Chapter outline: 29
2.1 Planning for audit 29
2.2 Audit standard 34
2.3 Resource mobilization and mapping 39
2.3 Audit methodology 41
Chapter 3: Measure level of performance and making improvements 46
Chapter description 46
Primary objective: 46
Enabling objectives: 46
Chapter outline 46
3.1 Data collection 46
3.2 Data Analysis and drawing conclusion 50
3.3 Making changes and improvements 53
3.4 Sustainability of changes 60
References 65
Annexes 66
Annex I: Clinical Audit proforma 66
Annex II: Gallery Walk 70
Annex III: Tips for successful development of clinical audit 71
Annex IV: Think-pair-share 73
Annex V: sample clinical audit for case study 74
Annex III: sample clinical audit for case study 74

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We are especially grateful to the Addis Ababa University Continuous Professional Development Directorate for their guidance and approval, ensuring the module meets the highest academic and professional standards. Additionally, we acknowledge the invaluable contributions of healthcare professionals and educators who provided practical feedback during the review process.

Finally, we thank all those who supported this initiative, directly or indirectly, in advancing healthcare quality through effective clinical audit training. Your commitment to excellence has been instrumental in bringing this important resource to execution.

List of tables and figures

List of Tables

Table 1: Chapter/session outline summary 13
Table 2: Training schedule 15
Table 3: comparison between clinical audit, research and service evaluation 24
Table 4: Stalk holder analysis framework 58
Table 1 - Criteria and standards ..... **Error! Bookmark not defined.**

List of figures

Figure 1: Cycles of clinical audit 28
Figure 2: Sources of clinical audit topic 30
Figure 3: Evidence pyramid (adopted from Tuffs university) 37
Figure 4: Fish-bone diagram 56
Figure 5: Forcefield analysis diagram 59

Forward

A clinical audit is a process used to evaluate and improve the quality of care provided to patients. It involves analyzing patient data and identifying areas where improvements can be made, and then implementing changes to address those areas. The goal of a clinical audit is to ensure that patients receive safe, effective, and efficient care. The results of a clinical audit can be used to inform practice and policy changes, as well as to identify areas for further research. The process of clinical audit can be forward by having a clear aim, selecting appropriate sample, data collection, data analysis, feedback, and implementation of changes.

Approval statement

As the Director of Continuous Professional Development (CPD) at Addis Ababa University, it is my privilege to endorse the Clinical Audit Continuous Professional Development Training Module. This manual represents a comprehensive and meticulously developed resource aimed at equipping healthcare professionals with the knowledge, skills, and competencies essential for conducting effective clinical audits. The training module aligns with our country's commitment to advancing evidence-based practices and fostering a culture of continuous improvement in healthcare delivery.

The manual has been carefully reviewed and approved for its academic rigor, practical relevance, and alignment with national and institutional priorities. Its structured approach, incorporating core competencies, systematic methodologies, and practical tools, ensures that participants can confidently plan, implement, and sustain impactful clinical audits in their respective settings. By addressing existing gaps and promoting measurable improvements in healthcare quality, this CPD module stands as a vital resource for enhancing patient outcomes and strengthening health systems.

I commend the Ethiopian Association of Anesthetists and all contributors for their dedication to developing this training manual, and I am confident that it will serve as an invaluable guide for advancing clinical audit practices across Ethiopia.

Dr. Merahi Kefyalew

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Abbreviations and acronyms

CC Core Competency

CCD Chief Clinical Director

CPD Continuous Professional Development

CEU Continuing Education Unit

TOT Training of Trainers

Introduction to the manual

This clinical audit training is a five days duration training course designed to equip health professionals and healthcare administrators involved in clinical services and working in different areas of health facilities aiming to improve efficiency of clinical service and increased patients' satisfaction on health care delivery.

"Clinical audit can be described as a quality improvement cycle that involves measurement of the effectiveness of healthcare against agreed and proven standards for high quality, and taking action to bring practice in line with these standards so as to improve the quality of care and health outcomes."

Clinical Audit forms the system for improving standards of clinical practice. Aspects of patient care are evaluated against expected standards of care and where necessary, changes are made at an individual, team or service level. A re-audit can then be used to confirm that improvements have been effective. Topics for clinical audit should reflect national and/or local priorities of areas of concerns.

Clinical audit is one of the methods to improve clinical effectiveness and improves patient safety. It is the process of systematic assessment of clinical service against the preset standard. Clinical audit is also considered as the key component of clinical governance. Clinical audit is a tool which can be used to realize how the clinical care is being provided against the explicit criteria and to learn if there are opportunities for improvement. Clinical audit may be used to improve aspects of care in a wide clinical area. It can also be used in association with changes in care provision or to confirm that current practice meets the expected level of performance.

Clinical audit has many advantages:

- It offers a way to assess and improve patient care
- Helps to uphold professional standards
- Help healthcare staff to identify and measure areas of risk within their service
- Helps to create a culture of quality improvement in the clinical setting
- Used as educational platform for the participants
- It involves being up to date with evidence based good practice
- It offers an opportunity for increased job satisfaction
- It is increasingly seen as an essential component of professional practice
- It can improve the quality and effectiveness of healthcare service

Scope of the manual

The aim of this document is to prepare organized training manual for healthcare providers and healthcare service administrators in Ethiopia to enable them aware of clinical audit concept, process of clinical audit, support best process in clinical audit, improve awareness of clinical audit as an integral part of clinical practice. Thus, this training course manual will provide practical guide to the methodological aspect of clinical audit including the wider considerations for ethical/data protection/confidentiality primarily for healthcare professionals and healthcare service administrators involved in clinical service currently and used as a reference during and after training delivered.

It is trusted that the module is clinically relevant and will improve the practice of clinical service. It is educationally all-encompassing, recognizing the specific needs of the adult learner. In keeping with this, it will comprise a range of teaching modalities to include skill sessions, scenarios, demonstrations, discussions and interactive lectures.

Core competency of the training

- After completion of this module, the participants will be able to conduct clinical audits on particular service areas in order to improve healthcare delivery

Rationale of clinical audit training (existing gaps and why addressing the gaps is important)

There is a considerable quality gap in the delivery of health care along one or more quality dimensions, including people-centeredness, safety, timeliness, effectiveness, efficiency, equity, and integration, according to a growing body of research. A well-organized, efficient clinical auditing program must be implemented as one of the measures taken to increase the quality and safety of the care given.

In order for clinical audit programs in healthcare facilities to be successful, two requirements must be met: the use of appropriate methodologies and creating a supportive environment.

As a result, there are problems with the application of appropriate methods in terms of careful planning, creating simple and practical audit standards and criteria, designing and monitoring appropriate quality improvement plans based on identified gaps, connecting audit findings with quality improvement projects (systematic management of change), as well as the absence of an appropriate structure that can organize and provide the necessary support for auditors in developing their audit project.

This document describes the idea of clinical audit, the procedures required in performing a clinical audit, the roles and duties of key participants, and techniques for monitoring the advancement of clinical audit projects. Additionally, to make it easier to conduct clinical audits regularly in health facility, simpler clinical audit tools on certain topic areas based on the clinical guidelines and protocols.

Module syllabus

1. **Module name**: Clinical audit
2. **Module code**: CAERBC 212
3. **Module duration**: 4 days
4. **Pre-requisite:** None
5. **Module continuing education unit (CEU)-** 15 CEU
6. **Module description**: This is a four days training module/course aimed at identifying areas of health care service and improving it through clinical audit, which is created for clinicians, policymakers, healthcare service leaders, and administrators. It focuses on the health service quality improvement activities.
7. **Module/course goal (CC)**: After completion of this module, participants will be able to conduct clinical audits on particular service areas in order to improve healthcare delivery.
8. **Enabling/learning objectives**

At the end of this training the trainees will be able to:

- Identify principles of clinical audit
- Plan to conduct clinical audit
- Identify best practice (standards)
- Mobilize and map resources to conduct clinical audit
- Measure level of performance
- Manage changes for healthcare delivery and maintain sustainability

9. **Chapters/sessions outlines**

Table 1: Chapter/session outline summary

Chapter	Title /topic	Durations	Sessions
1.	Principles of clinical audit	5½ Hours	Session I: introduction to clinical audit Session II: clinical audit cycle
2.	Clinical audit proforma	8½ hours	Session I: Planning for audit Session II: Audit standard Session III: Resource mobilization and mapping Session IV: Audit methodology

2	Measure level of performance and making improvements	10½ Hours	Session I: Data collection Session II: Analysis of data and drawing conclusion Session III: Making changes Session IV: Sustainability of changes

10. Teaching learning methods and activities

- Interactive lectures
- Small Group work, reporting and feed back
- Role play
- Case studies
- Guided activity of using online resources
- Feedback on self-reflection
- Gallery walks
- Question and answer

11. Learning materials and resource (including references)

Training materials

- Lap top computer
- LCD projector
- White board
- White board marker
- Flip chart
- Permanent marker
- Note book
- Pen
- Sticker notes

12. Learning Assessment methods

- Written exam
- Observation of case study analysis in small groups
- Observation of group activity/ audit proposal presentation
- Review of the completed clinical audit document

13. Criteria for course completion and Criteria for certification /earning CEU

To earn 15 CEU after completion of training, the trainees must score 80 and above points from the following assessments.

- Written test (20%)
- Observation of small group case study analysis (10%)

- Observation of group presentation of clinical audit proposal (20%)
- Review of completed individual clinical audit document (50%): This should be conducted as a home take project after the first phase of the training and submitted on the agreed time period set during the training or at the end of the training based on the ready-made data given to the trainees.

14. Trainer and facilitators selection criteria

- Have basic/TOT on clinical audit training or course
- Participate on clinical audit training module preparation

15. Trainees and participants selection criteria

- All healthcare professionals currently engaged in clinical service
- Health care administrators/leaders and healthcare policy makers

16. Training evaluation methods (satisfaction, learning, behavior and impact)

- Trainees' reaction using daily evaluation and end of training evaluation
- Direct observation of trainees' behaviors
- Interview of trainees and facilitators

17. Venue:

Accredited CPD provider center

18. Composition

Maximum of 20 participants from health institution (teaching/non-teaching)

Physicians, anesthetists, pharmacists, healthcare leaders (CCD, Quality directors)

Number of trainers: 4, number of facilitators: 3

19. Schedule

Table 2: Training schedule

Time	Day 1	Day 2	Day 3	Day 4
8:30-4:30	Registration Pretest Chapter 1.1: Introduction to clinical audit	Discussion on day one sessions Chapter 2.1 Planning for audit	Chapter 2.4: Audit methodology continued	Chapter 3.2 data analysis and drawing conclusion continued Chapter 3.3 making changes and improvement

Tea-break				
10:45-12:30	Chapter 1.1 Introduction to clinical audit continued	Chapter 2.1 Planning for audit continued Chapter 2.2: audit standard	Chapter 3.1: Data collection	Chapter 3.3: making changes and improvement continued Chapter 3.4: sustaining improvement
Lunch				
2:00-3:30	Chapter 1.2 clinical audit cycle	Chapter 2.2: audit standard continued	Chapter 3.1: Data collection continued Chapter 3.2 data analysis and drawing conclusion	Chapter 3.4: sustaining improvement continued
Tea break				
3:45-5:30	chapter 1.2 Clinical audit cycle continued	Chapter 2.3 resource mobilization and mapping Chapter 2.4: Audit methodology	Chapter 3.2 data analysis and drawing conclusion continued	Post test Closing

Chapter 1: principles of clinical audit

Duration: 5½ Hours

Chapter description

This chapter is focused on the over view of key concepts, principles and types of clinical audit and audit cycle. It outlines the basic definition of clinical audit, comparison of clinical audit with research and service evaluation. It also describes the basic stages of audit cycles.

Primary objective:

At the end of this chapter, participants will be able to analyze principles of audit and audit cycle to conduct the quality improvement clinical audit.

Enabling objectives:

At the end of this session, participants will be able to:

- Describe clinical audit
- Describe the importance of conducting clinical audit
- Discuss the difference between clinical audit, research and service evaluation
- Explain types of clinical audit
- Evaluate the stages of audit cycles

Chapter outline

Outline

1.1 Introduction to clinical audit

1.2 Audit cycle

1.1 Introduction to clinical audit

Session duration: 170 minutes

Session specific objectives: At the end of this session, participants will be able to define clinical audit, describe the scope of clinical audit, and explain types of clinical audit.

Activity 1: Question and answer

- What is Clinical Audit?
- What types of clinical audit/quality improvement projects do you know?

- What is the difference between clinical audit, research and service evaluation

Time: 15 minutes

Historical background of clinical audit

Clinical audit was started back in the mid 19 century During the Crimean War of 1853–1855. The unsanitary conditions and high troop mortality rates in the hospital alarms miss Florence Nightingale and her colleges. She implemented strict sanitary procedures and hygiene standards for the facility and equipment along with her team of 38 nurses. Miss Nightingale, who was gifted in arithmetic and statistics, started keeping thorough records of the hospital patients' mortality rates.

Following the change of regime, the mortality rates fell from 40% to 2% and were instrumental in getting Florence's procedures accepted by the Officers and Doctors. Her methodical approach, as well as the emphasis on uniformity and comparability of results of healthcare, is recognized as one of the earliest programs of outcomes management.

Ernest Codman, (1869-1940) was another advocator of clinical audit through his work on monitoring of surgical outcomes in 1912 and recognized as the first true medical auditor.

Despite the successes of Florence Nightingale and Ernest Codman, clinical audit was slow to catch on. Only a small minority of healthcare staff embraced the process as a means of measuring the quality of care delivered to patients for the next 130 years.
Throughout the years the concept and definition of clinical audit has been evolved from 20th century medico-centric view to the multi-disciplinary approach which is patient-centered approach. The changes can be seen from the following definition of clinical audit.

"**Working for Patients, 1989"** – This White Paper defined medical audit, (as it was then known), as "the systematic critical analysis of the quality of medical care including the procedures used for diagnosis and treatment, the use of resources and the resulting outcome and quality of life for the patient."
Medical audit later evolved into clinical audit and a revised definition was announced, "clinical audit is the systematic analysis of the quality of healthcare, including the procedures for diagnosis, treatment and care, the use of resources and resulting outcome and quality of life for the patient." ***(NHS Executive 1993)***
The National Institute for Health and Clinical Excellence, **(NICE**), published "The Principles for Best

Practice in Clinical Audit in 2002, which defined clinical audit as, "a quality improvement process that seeks to improve patient care and outcomes through a systematic review against explicit criteria and the implementation of change. Aspects of the structure, processes and outcomes of care are selected and systematically evaluated against explicit criteria. Where indicated, changes are implemented at an individual, team or service level and further monitoring is used to confirm improvement to healthcare delivery."

Nevertheless all these definitions are intended to embody three key attributes:

• Recognisably high standards of care

• Transparent responsibility and accountability for those standards and

• A constant dynamic of improvement

The current concept of clinical audit encourages the involvement of patients. **Iain Thomas, Myocardial Ischemia Audit Project, (MINAP),** Lay Representative is quoted as saying "Clinical audit must include patients. They, as well as the clinicians, are the true professionals in illness. Although the clinicians are well-qualified to make a diagnosis and prescribe a course of action, only the patient truly understands the stress and physical and emotional suffering that come with having a disease. The patient's perspective is a crucial component of audit."

Furthermore, it is important that clinical audit is supported by those who have the authority and commitment to see changes put into practice It is increasingly understood that without patient involvement, services cannot be enhanced. **In the words of the White Paper, "Liberating the NHS", (2010),** "**nothing about me without me".**

Activity 2: Group activity:

Be in group of 3-5 discuss the importance of conducting clinical audit reflect to the larger group

Time: 15 minutes

Importance of clinical audit

- Clinical audit offers a way to assess and improve patient care, to uphold professional standards and 'do the right thing'.

- Through clinical audit, healthcare staff may identify and measure areas of risk within their service.
- Regular audit activity helps to create a culture of quality improvement in the clinical setting.
- Clinical audit is educational for the participants. It involves being up to date with evidence based good practice.
- It offers an opportunity for increased job satisfaction
- It is increasingly seen as an essential component of professional practice
- It can improve the quality and effectiveness of healthcare
- Builds relationships between clinicians, clinical teams, managers and patients

- Advantages of involvement in Clinical Audits:
  - Shows patients that you reflect on and improve your care
  - Provides evidence that you have met training requirements
  - Demonstrates commitment to progress in your career
  - Distinguishes you from your colleagues in a competitive situation
  - Win prizes in audit competitions
  - **Publications**
  - **Provide links with research**

Types of Clinical Audit

There are different types of clinical audit.

- Processed based audit
- Outcome based audit
- Structural audit
- Comparative audit
- Benchmarking audit
- Clinical governance audit
- Patient satisfaction

The commonest types of clinical audits are process audit and outcome-based audit.

Process-based audit is a type of clinical audit that focuses on the processes used to deliver healthcare services. It looks at the steps taken to provide care and assesses whether they are being done as per the agreed upon standard. This type of audit is used to identify areas where processes can be improved or streamlined to improve the quality of care. It can also be used to identify any potential risks or hazards that may be present in the delivery of care.

Activity 3: **Think-Pair-Share**

Provide process based clinical audit topics

Instruction

- Think about the clinical service in your institution and identify process based clinical audit topic share in pair with participant next to you. Think for 2 minutes by your own and share in pair for 3 minutes.
- Reflect your idea and discuss in larger group/whole class.

Time: 15 minutes

Process-based clinical audit topics can be used to assess the effectiveness of processes and procedures in place to ensure that the highest standards of care are being met. One process-based clinical audit topic could be the assessment of patient safety protocols. This could involve evaluating the processes and procedures in place to ensure that patients are receiving the safest care possible. This could include assessing the use of safety protocols such as **hand hygiene, proper use of personal protective equipment, and the use of safety checklists**. It could also involve evaluating the effectiveness of patient education and communication processes to ensure that patients are aware of the risks associated with their care and are taking the necessary steps to reduce those risks.

Another process-based clinical audit topic could be the assessment of clinical documentation. This could involve **evaluating the accuracy and completeness of clinical documentation**, as well as the **timeliness of documentation**. It could also involve assessing the use of standardized forms and templates to ensure that all relevant information is being captured in a consistent manner.

A third process-based clinical audit topic could be the assessment of medication management processes. This could involve **evaluating the accuracy and completeness of medication orders**, as well as the

timeliness of medication administration. It could also involve assessing the use of standardized forms and templates to ensure that all relevant information is being captured in a consistent manner.

Overall, process-based clinical audit topics can be used to assess the effectiveness of processes and procedures in place to ensure that the highest standards of care are being met. By evaluating these topics, healthcare organizations can identify areas for improvement and ensure that they are providing the highest quality of care to their patients.

Outcome-based audit is a type of clinical audit that focuses on the outcomes of healthcare services. It looks at the results of care and assesses whether they are meeting the desired goals. This type of audit is used to identify areas where outcomes can be improved or where additional resources may be needed. It can also be used to identify any potential risks or hazards that may be present in the delivery of care.

Activity 4: Think-Pair-Share

Provide outcome based clinical audit topics

Instruction

- Think about the clinical service in your institution and identify process based clinical audit topic share in pair with participant next to you. Think for 2 minutes by your own and share in pair for 3 minutes.
- Reflect your idea and discuss in larger group/whole class.

Time: 15 minutes

Outcome based clinical audit topics can be used to measure the effectiveness of care and to identify areas for improvement.

Outcome based clinical audit topics can focus on a variety of areas, including patient safety, patient satisfaction, clinical outcomes, and cost-effectiveness. For example, an audit could focus on the effectiveness of a particular medication or treatment, or the accuracy of a diagnosis. It could also focus on the quality of care provided by a particular healthcare provider or facility.

When selecting outcome based clinical audit topics, it is important to consider the goals of the audit. For example, if the goal is to improve patient safety, then the audit should focus on areas such as medication errors, infection control, and patient falls. If the goal is to improve patient satisfaction, then the audit should

focus on areas such as communication between healthcare providers and patients, wait times, and patient education.

It is also important to consider the resources available for the audit. For example, if the audit is to be conducted in a hospital setting, then the resources available may include patient records, medical staff, and equipment. If the audit is to be conducted in a community setting, then the resources available may include patient surveys, community health workers, and health education materials.

Finally, it is important to consider the timeframe for the audit. Depending on the goals of the audit, the timeframe may range from a few weeks to several months. It is important to ensure that the timeframe is realistic and achievable, and that the audit is conducted in a timely manner.

Overall, outcome based clinical audit topics are an important part of healthcare quality improvement. By selecting the right topics and resources, and setting a realistic timeframe, healthcare providers can ensure that the audit is conducted effectively and efficiently.

Both types of clinical audit are important for assessing the quality of healthcare services and identifying areas for improvement. Process-based audit is used to identify areas where processes can be improved or streamlined, while outcome-based audit is used to identify areas where outcomes can be improved or where additional resources may be needed. Both types of audit can be used to identify any potential risks or hazards that may be present in the delivery of care.

Difference between Clinical Audit, Research and Service Evaluation

Activity 5: gallery walk

- Make the class in three groups for this activity.
- Take one of the topics: Clinical Audit, Research and Service Evaluation
- Discuss in your group about its purpose, objective, final result on flip chart.
- The flip chart of all groups will be posted in different areas of the class and one participant from each group form other group at each flip chart posts/gallery
- Each participant will brief at the post will brief his/her group ideas for other groups members, receive feed backs and move to the next flip chart post other do the same thing.

Time: 30 minutes

Clinical audit, research and service evaluation are all important tools for improving healthcare services. Each of these approaches has its own unique purpose and benefits, and understanding the differences between them can help healthcare professionals make informed decisions about how to best improve their services. The difference is based on their aim, scope, duration and methodology.

Clinical audit is a process of systematically reviewing the quality of care provided to patients. It involves collecting data on the care provided, analyzing it, and then making recommendations for improvement. Clinical audit is used to identify areas of practice that need improvement, and to ensure that the care provided is of the highest quality.

Research is a systematic process of collecting and analyzing data to answer questions about a particular topic. Research can be used to identify new treatments or interventions, to evaluate existing treatments or interventions, or to understand the causes of a particular health problem. Research is often used to inform policy decisions and to improve healthcare services. Service evaluation is a process of assessing the quality of a healthcare service. It involves collecting data on the service, analyzing it, and then making recommendations for improvement.

Service evaluation is used to identify areas of practice that need improvement, and to ensure that the service is of the highest quality. In summary, clinical audit, research and service evaluation are all important tools for improving healthcare services. Each of these approaches has its own unique purpose and benefits, and understanding the differences between them can help healthcare professionals make informed decisions about how to best improve their services.

Table 3: comparison between clinical audit, research and service evaluation

Clinical audit	Research	Service evaluation
Based on facts, (standards)	Aims to establish what expected practice is	Designed and conducted solely to define or judge current care
Aims to evaluate how close practice is to expected practice	Is often a one off study	Identifies the standards that the service achieves
Is specific & local to one particular patient group, (results are not transferable to other settings)	Is designed so it can be replicated and results generalized to other similar groups	Measures current service without reference to a standard

Aims to improve services	Aims to generate new knowledge or increase the sum of knowledge	Usually involves analysis of existing data, but may include administration of simple interview or questionnaire
Is practice based	Is usually initiated by researchers	
Never involves a new treatment.	Is theory driven	-
-	Is usually testing a hypothesis and follows a protocol	-
May require Ethical review	Requires Ethical review	-

Do you want to measure current practice against evidence based clinical standards? If the answer is yes – It is probably ***Clinical Audit***

Do you want to investigate the effect of a new or existing treatment or technique on patients/carers? If the answer is yes – It is probably ***Research***

Do you want to evaluate the effectiveness and/or efficiency of current practice or service? If the answer is yes – It is probably S***ervice Evaluation***

Summary

Clinical audit is a process that has been defined as "a quality improvement" process that seeks to improve patient care. through systematic review of care against explicit criteria and the implementation of change. And further monitoring is used to confirm improvement in healthcare delivery.

Clinical audit is a process that involves reviewing and evaluating the quality and safety of healthcare services provided to the patients. It is conducted by healthcare professionals and aims to identify areas for improvement in order to enhance patient care and outcomes. It typically includes the systematic review of medical records, patients' satisfaction surveys, and interviews. The findings of audit then will be used to develop and implement changes. The goal of clinical audit is to improve quality of patient care by identifying areas of weakness, implementation changes and monitoring to ensure that they are effective.

1.2 Audit cycle

Session duration: 160 minutes

Session specific objectives: at the end of this session, participants will be able to describe cycles of clinical audit.

Group Activity: answer and question
- What are the stages of audit cycles?
Time: 10 minutes

An audit cycle refers to the process of conducting an audit, which typically includes planning, conducting, and reporting. The clinical audit cycle typically consists of four stages: planning and preparation, implementation, evaluation, and follow-up.

Planning and preparation

The planning phase involves determining the scope of the audit and developing an audit plan. During the planning stage, the audit team will develop an audit plan that outlines the objectives, scope, and methodology of the audit. This plan will also include the criteria that will be used to evaluate the services and processes being audited. It also includes available resource identification, topic selection, stakeholder involvement and planning for the delivery of audit field work.

Set standard /criteria

Standards should be 'robust' and evidence based

Useful sources for standards include**:** local standards in the form of evidence based guidelines; nationally endorsed clinical guidelines; standards and clinical guidelines from relevant quality and safety programs

If national or local guidelines are not available, a literature review may be carried out to identify the best and most up to date evidence from which audit criteria may be generated

Define the criteria - this should be in the form of a statement, e.g. all patients with hypertension who smoke should be offered smoking cessation advice

Define the standard - usually a target (percentage) this may be a minimum standard or an optimal one, depending on the scenario.

Measure level of performance

The implementation stage involves the actual audit process. During this stage, the audit team will collect data and analyze it to identify areas of improvement. The team will also review the policies and procedures in place to ensure that they are in line with best practices.

Making improvement

Present the results and discuss them with the relevant teams in the organization

• The results should be used to develop an action plan, specifying what needs to be done, how it will be done, who is going to do it and by when The team will also provide feedback to the healthcare organization on the areas that need improvement.

Follow up/sustain improvement

This follows up the previous stages of the audit, to determine whether the actions taken have been effective, or whether further improvements are needed

• It involves repeating the audit (i.e. targets, results, discussion); hence the terms “audit cycle”.

The follow-up stage is when the healthcare organization will implement the recommendations made by the audit team. This stage is important to ensure that the improvements are being made and that the healthcare organization is meeting the standards of care. Clinical audit cycles are an important part of healthcare quality assurance. They help to ensure that healthcare services are being provided in a safe and effective manner and that best practices are being followed.

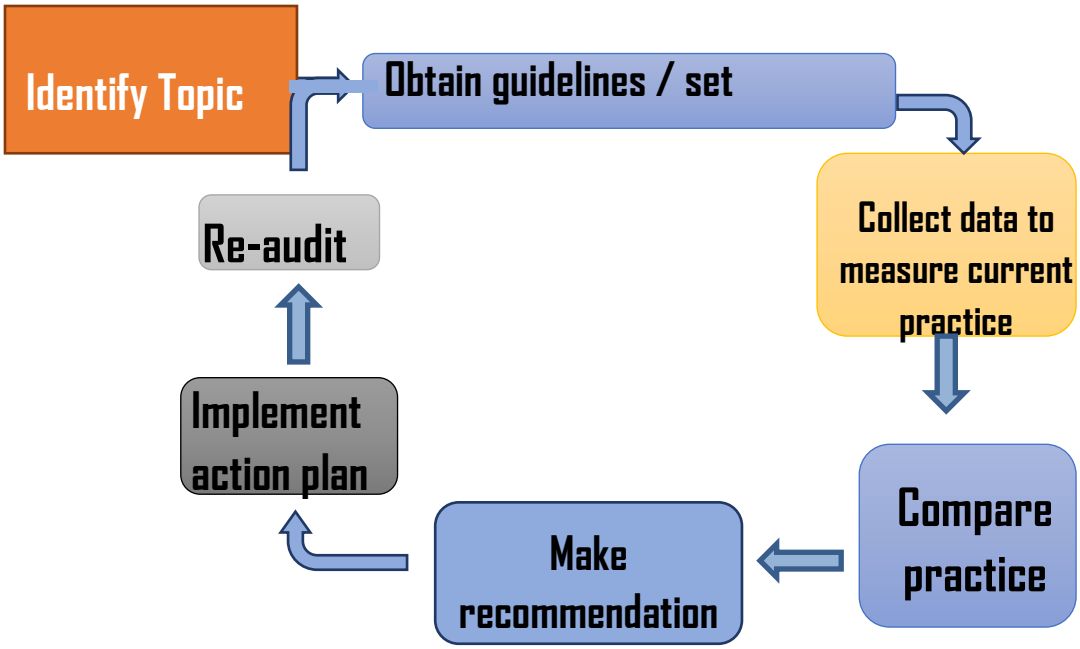

Figure 1: Cycles of clinical audit

Summary

The audit cycle is a process of conducting an audit which includes planning, implementation, evaluation, and follow-up. Clinical audit cycles are important for healthcare quality assurance, ensuring that services are provided safely and effectively and that best practices are followed.

Chapter two: Clinical audit proforma

Duration: 8 ½ Hours

Chapter description:

This chapter is designed to focus on how to develop an organized clinical audit proposal on selected service area. It mainly describes clinical audit planning process, audit methodology, audit objectives and an audit tool. It also includes identifying a team to conduct the audit, communicating the plan to all relevant stakeholders.

Primary objective:

At the end of this chapter participants will be able to demonstrate comprehension of the audit planning process, audit methodology and clinical audit tool.

Enabling objectives:

At the end of this session, participants will be able to:

- Select appropriate audit topic based on topic selection criteria
- Set appropriate clinical audit aim and objectives
- Analyze the audit standard by identifying best practices in the area
- Construct audit methodology using audit planner
- Develop organized clinical audit tool

Chapter outline:

2.1 Planning for audit

2.2 Audit standard

2.3 Resource mobilization and mapping

2.4 Audit methodology

2.1 Planning for audit

Session duration: 160 minutes

Session specific objectives: At the end of this session, participants will be able to determine prioritized service area for clinical audit with appropriate aim and objective.

Activity 1: question and answer
Think and write criteria for audit topic selection
Time: 10 minutes

Topic selection is a crucial stage that has to be carefully considered. Clinical audit subjects should be chosen with the goal of enhancing the standard of care or safety of service delivery. The possible sources of clinical audit project may include publication of conclusive **new evidence** about clinically effective healthcare, Local or regional treatment **guidelines, protocols or frameworks**. These will ideally be based on best evidence (perhaps from national guidelines), or maybe a local consensus of best practice. Other sources include **User views or complaints**, Adverse incident/near miss reporting, aka **clinical/critical incident reporting**, identified local priorities or concerns e.g. **areas of high volume, risk or cost.** Examples of audit topics include:

- Does the patients intraoperative record complete?
- Have patient history and examination been properly recorded following admission?
- Were appropriate investigations carried out?
- Was the treatment appropriate and timely?

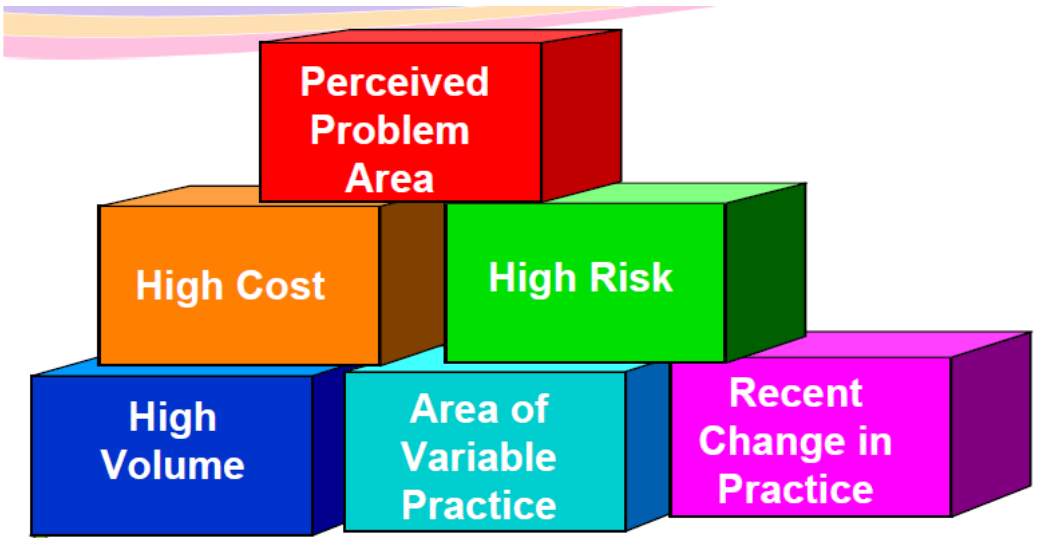

Figure 2: Sources of clinical audit topic

Activity 2: Group activity

1. be in a group 5 participants and select three topics for clinical audit with respect to your clinical service area
2. Discuss why it is important for clinical audit.
3. Present your topic and discuss in larger group with justification. The other group members will give feed-back.
4. Summarize the activity with facilitator

Time: 30 minutes

Prioritizing possible audit topics

Selection of the audit topic needs careful thought and planning, as clinical staff and service providers have limited resources with which to deliver clinical audits. Mandatory audits will take resource priority. All other audits should therefore be prioritized to ensure that available resources are used effectively. These audits should focus on areas with the greatest need to improve practice.

Clinical audit topics are typically chosen based on several criteria, including:

1. **Relevance:** The topic should be relevant to the care provided by the healthcare organization and should address a specific area of clinical practice or patient care.

2. **Importance:** The topic should be important in terms of improving patient outcomes, reducing errors or adverse events, or increasing efficiency and cost-effectiveness.

3. **Feasibility**: The topic should be feasible to audit, with clear and measurable outcomes and a clear plan for data collection and analysis.

4. **Data availability:** There should be adequate data available to conduct the audit, such as patient records, clinical guidelines, and other relevant information.

5. **Prioritization:** The topic should be prioritized based on the organization's goals and objectives, as well as feedback from staff and patients.

6. **Compliance:** The topic should ensure the organization is in compliance with relevant laws, regulations, and accreditation standards.

A scoring system can be used to prioritize audit topics in order of importance. An example of a simple scoring system is where each topic is given a score between 1 and 5 based on importance (with 1 being the lowest and 5 being the highest) on two measures:

(i) The impact of a resulting continuous improvement plan on clinical effectiveness and quality of care.

(ii) The importance/urgency of the audit topic

Further examples of scoring systems for prioritizing audit topics may be found at:
http://www.hqip.org.uk/assets/Downloads/Clinical-Audit-Program-Guide-andGuidance-Tools.pdf

Activity 3: Group activity

1. In your previous group formed at beginning of the session, take your topics of each group identified as clinical audit and evaluate based on the topic selection criteria.
2. Present your topic in larger group and choose one audit topic based on the priority to develop audit proforma

Time: 30 minutes

Aim and objectives

Clinical audit is a process used to measure and improve the quality of care provided to patients. Establishing clear objectives for clinical audit is essential for ensuring that the audit is successful and that the results are meaningful. The purpose of the audit may be outlined in the form of aims and objectives.

AIM

Consider what you plan to accomplish when defining the goal of your clinical audit project. For example, the overarching goal shouldn't just be to "count the number of" or "examine," but should instead direct your audit toward accomplishing practice improvements where they are required. A declaration of what you intend to happen as a result of the audit can be included in the aim. Positive language should be used while making statements to ensure that the audit results in practice improvements.

OBJECTIVES

Your aim provides a broad structure for your clinical audit project. This may need to be broken down into a series of smaller objectives. Objectives are the steps that need to be taken in order to assess whether or not you have achieved your aim. Your objectives can be written as either specific tasks to be undertaken or as the different aspects of quality that your project will address. The objectives of a clinical audit should be

specific, measurable, achievable, relevant, and time-bound (SMART). This will help to ensure that the audit is focused and that the results are meaningful. When establishing objectives for a clinical audit, it is important to consider the purpose of the audit. Is it to identify areas of improvement, to ensure that care is in line with best practice, or to evaluate the effectiveness of a particular intervention? Once the purpose of the audit has been established, the objectives should be tailored to meet this purpose. It is also important to consider the scope of the audit. What areas of care will be included in the audit? What data will be collected and analyzed? What criteria will be used to evaluate the quality of care? These questions should be answered before the objectives are established.

Finally, it is important to consider the resources available for the audit. What resources are available to collect and analyze the data? What resources are available to implement any changes that may be recommended as a result of the audit?

Activity 4: Gallery-walk

1. Be in your previous group of 5 participants and develop aim and objective for the audit topic selected in your last session.
2. Write your Aim and Objective on flip chart
3. Reflect your aim and objective of clinical audit in larger group and discuss
4. Every group will give feedback

Time: 20 minutes

Summary:

The choice of a topic is an important step that requires careful thought. Given the limited resources available to clinical staff and service providers to conduct clinical audits, choosing the audit issue requires considerable consideration and planning. Clinical audit is a procedure used to evaluate and enhance the standard of patient care. To ensure that the clinical audit is successful and that the results are useful, clear objectives must be established.

2.2 Audit standard

Session duration: 120 minutes

Session specific objective: at the end of this session, participants will be able to determine level of evidence and construct agreed standard for clinical audit.

Activity 1: Group activity

With your group formed before

1. Describe source of evidences for clinical audit
2. Discuss level of evidence
3. Reflect your activity to larger group

Time: 20 minutes

The aim of clinical audit is to identify areas of practice that need improvement and to develop strategies to improve the quality of care. In order to ensure that clinical audit is effective, it is important to select and develop appropriate performance levels. Performance levels should be based on evidence-based standards and should be tailored to the specific needs of the organization.

Clinical audit standards and criteria are used to ensure that the care provided is of the highest quality and meets the needs of the patient. Clinical audit standards and criteria are developed by professional organizations, such as the Joint Commission on Accreditation of Healthcare Organizations (JCAHO) and the National Committee for Quality Assurance (NCQA). These standards and criteria are based on evidence-based practice and are designed to ensure that the care provided is safe, effective, and of the highest quality. The standards and criteria used in clinical audit are typically divided into two categories: process standards and outcome standards. Process standards are used to evaluate the processes used to provide care, such as the use of evidence-based practice, patient safety protocols, and communication between healthcare providers. Outcome standards are used to evaluate the results of care, such as patient satisfaction, clinical outcomes, and cost-effectiveness. The performance levels should be measurable and achievable, and should be regularly monitored and evaluated. When selecting performance levels, it is important to consider the context of the organization and the population it serves. The performance levels should be realistic and

achievable, and should be based on the best available evidence. For criteria to be valid and lead to improvements in service user care, they should be consistent with SMART guidance:

- Specific (explicit statements, Clear, unambiguous and jargon-free,).
- Measurable.
- Achievable (of a level of acceptable performance agreed with stakeholder).
- Relevant (related to important aspects of care,).
- Theoretically sound or timely (evidence based).

It is also important to consider the resources available to the organization, as well as the potential impact of the performance levels on patient outcomes. Once performance levels have been selected, it is important to develop strategies to ensure that they are met. This may include developing protocols and guidelines, providing training and education, and implementing quality improvement initiatives. It is also important to ensure that the performance levels are regularly monitored and evaluated, and that any necessary changes are made. Standards should be 'robust' and evidence based. Useful sources for standards include:

- local standards in the form of evidence-based guidelines;
- nationally endorsed clinical guidelines;
- standards and clinical guidelines from relevant quality and safety programs,
- clinical care programs and professional bodies; and
- Clinical guideline development organizations such as NICE…..

Tips in developing audit standards

- If standards are available in the form of guidelines or protocols, you should base your audit on the most widely applicable guidelines available, e.g. national guidelines rather than regional or local guidelines. However, it is important to bear in mind that guidelines are only as good as the

evidence they are based on. Even some national guideline statements are only 'good practice', without any research evidence base.

- If guidelines/ protocols do not exist, or existing ones are out of date, you will need to undertake a literature search to identify best practice.
- It is important that there is agreement with your standards locally before you start. You will find it hard to improve practice if there is disagreement as to what constitutes best practice.
- Standards should always be based on the strongest, most up-to-date, evidence of what constitutes best practice

The level of evidence in clinical audit refers to the strength and quality of the data used to support the findings and recommendations of the audit.

The levels of evidence in clinical audit include:

Level 1: Evidence from systematic reviews of multiple well-designed randomized controlled trials

Level 2: Evidence from a single well-designed randomized controlled trial

Level 3: Evidence from well-designed controlled trials without randomization

Level 4: Evidence from well-designed case-control and cohort studies

Level 5: Evidence from systematic reviews of descriptive and qualitative studies

Level 6: Evidence from a single descriptive or qualitative study

Level 7: Evidence from the opinion of experts, case reports, or standard of care.

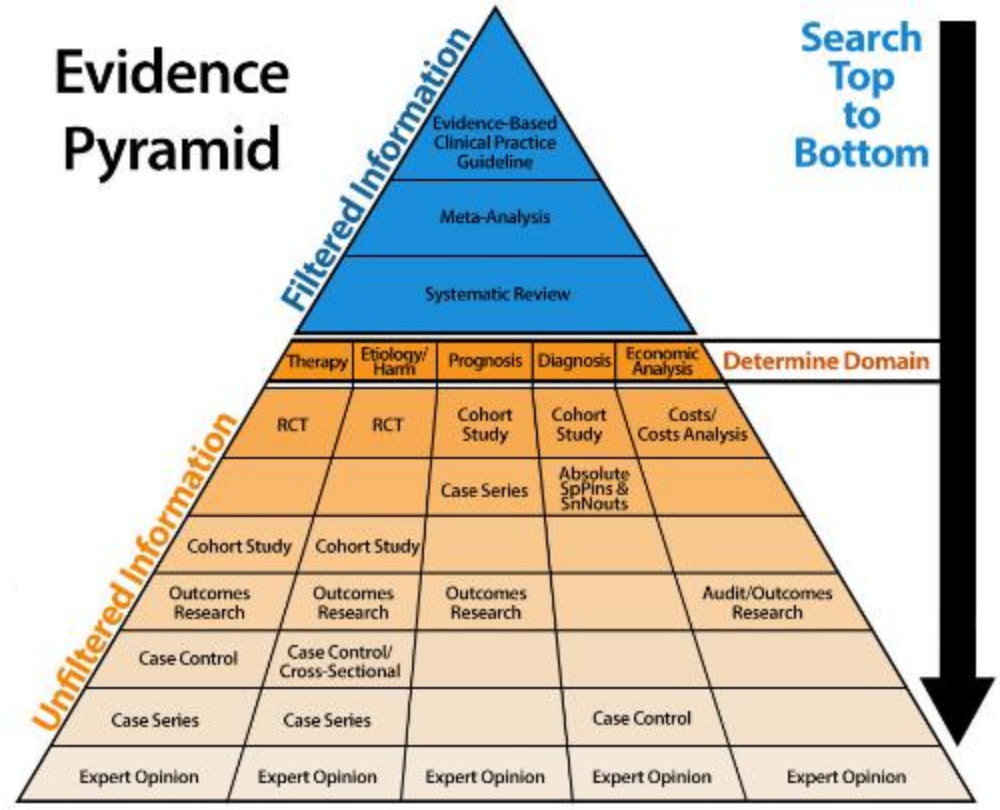

Figure 3: Evidence pyramid (adopted from Tuffs university)

Target level of performance

A Target/level of performance is a defined level or degree of expected compliance with audit criteria; may be expressed in percentage or proportion of cases.

Determine and develop appropriate performance levels

Audit criteria should consist of measurable statements of what should be happening with explicit and quantifiable performance levels. These performance levels or targets may be expressed as percentages. There are a number of ways to set targets for compliance, including discussion and development of a consensus opinion among audit team members and relevant stakeholders and benchmarking against national rates. Three factors should be taken into account and assessed when setting targets. These factors are clinical importance, practicability and acceptability. The expected level of performance or target can range from 0% (the criterion is something that must never be done) to 100% (the criterion is something that must always be adhered to).

Clinical Importance

Where a criterion is critical to the safety of service users, targets may be set at 100% or 0%, for example, a clinical audit relating to safe administration of medication could have a target of 100% for the following criterion 'medication is not administered to a service user with a known allergy to the medication'

Practicability

However, where clinical importance is not as significant, resources required to fulfill the target performance level should be considered and an acceptable performance level (one which is seen as both reasonable and attainable by those delivering and receiving care) should be identified, for example, in a clinical audit relating to the time frame within which service users should be seen in a particular outpatient clinic, a target of 90% may be deemed appropriate.

Acceptability

An optimum level of performance is set when the best care possible is identified given the resources available and normal conditions of care-giving. This will lie somewhere between the minimal acceptable level of care and the highest possible level of care (possible under ideal conditions, with no restrictions on resources).

Activity 2: guided practice

With your group formed previously

1. Demonstrate evidence search from different data bases
2. Filter standards/evidences for your Audit Topic

Time: 30 minutes

Summary

Setting standards for clinical audit involves establishing guidelines and criteria for evaluating the quality of care and identifying areas for improvement. This can include measures such as patient outcomes, adherence to best practices, and patient satisfaction. The standards for clinical audit should be evidence-based and regularly reviewed and updated to ensure they remain relevant and effective. Standard setting should involve input from healthcare professionals, patients, and other stakeholders to ensure that the standards are relevant and meaningful.

2.3 Resource mobilization and mapping

Session Duration: 60 minutes

Session specific objective: At the end of this session, participants will be able to:

- Identify resources required need needed to improve service delivery through clinical audit.
- Mobilize and allocate resources required need needed to improve service delivery through clinical

Activity 1: think-pair-share

- Consider a service area in your institution which needs improvement and think about how do you identify the required resource and mobilize from the stalk holders by your own for 3 minutes.
- Share your ideas in pair with participant next to you and discuss
- Reflect your pair shared ideas and discuss in larger group

Time: 15 minutes

Resource identification

Resource identification is the process of identifying and labeling resources in order to make them easier to locate and manage. It is an important part of the overall resource management process, as it helps to ensure that resources are properly identified and tracked. Resource identification is typically done by assigning a unique identifier to each resource, such as a serial number, a name, or a code. This identifier can then be used to quickly locate and manage the resource.

Resource identification is important for a variety of reasons. First, it helps to ensure that resources are properly tracked and accounted for. This is especially important in organizations that have multiple resources, as it helps to ensure that all resources are properly identified and accounted for. Additionally, resource identification helps to ensure that resources are properly allocated and used. By assigning a unique identifier to each resource, it is easier to track and manage the resource, as well as to ensure that it is being used in the most efficient manner.

Resource identification is also important for security purposes. By assigning a unique identifier to each resource, it is easier to track and monitor the resource, as well as to ensure that it is not being used for malicious purposes. Additionally, resource identification helps to ensure that resources are properly protected from unauthorized access. By assigning a unique identifier to each resource, it is easier to ensure that only authorized personnel have access to the resource.

Overall, resource identification is an important part of the overall resource management process. By assigning a unique identifier to each resource, it is easier to track and manage the resource, as well as to ensure that it is being used in the most efficient manner. Additionally, resource identification helps to ensure that resources are properly protected from unauthorized access.

Allocation of resources

Resource allocation is the process of assigning resources to tasks or activities in order to achieve a desired goal. It is a critical part of project management and involves the careful consideration of available resources, such as time, money, personnel, and materials, and how they should be used to best meet the project's objectives. Resource allocation is a key factor in determining the success of a project, as it ensures that the right resources are available when and where they are needed.

Resource allocation is a complex process that requires careful planning and consideration. It involves analyzing the project's objectives and determining the resources needed to achieve them. This includes assessing the availability of resources, such as personnel, materials, and funds, and determining how they should be allocated to best meet the project's goals. It also involves making decisions about how to prioritize tasks and activities, and how to allocate resources to them.

Resource allocation is an iterative process that requires ongoing monitoring and adjustment. As the project progresses, resources may need to be reallocated to different tasks or activities in order to ensure that the project is on track and that the desired outcomes are achieved. It is also important to consider the impact of resource allocation decisions on other projects and activities, as well as the potential for resource conflicts.

Resource allocation is a critical part of project management and requires careful planning and consideration. It is important to ensure that the right resources are available when and where they are needed, and that they are allocated in a way that best meets the project's objectives. By carefully managing resource allocation, project managers can ensure that their projects are successful and that the desired outcomes are achieved.

Resource mobilization:

Resource mobilization is a term used to describe the process of gathering resources to achieve a specific goal. It is a key component of any successful project or organization, as it involves the identification,

acquisition, and utilization of resources to achieve a desired outcome. Resource mobilization can involve a variety of resources, including financial, human, and material resources.

Financial resources are the most common type of resource mobilized, as they are necessary to fund projects and operations. This can include grants, donations, and other forms of funding. Human resources are also important, as they are necessary to carry out the project or organization's activities. This can include volunteers, staff, and other personnel. Material resources are also necessary, as they are necessary to carry out the project or organization's activities. This can include equipment, supplies, and other materials.

Resource mobilization is a complex process that requires careful planning and coordination. It involves identifying the resources needed, assessing their availability, and then acquiring them. It also involves utilizing the resources in an efficient and effective manner. This can include developing strategies to maximize the use of resources, as well as ensuring that resources are used in a way that is consistent with the project or organization's goals.

Resource mobilization is an important part of any successful project or organization. It requires careful planning and coordination, as well as the identification, acquisition, and utilization of resources. By mobilizing resources in an efficient and effective manner, organizations and projects can achieve their desired outcomes.

Summary

Resource mobilization and mapping in clinical audit is an important part of the audit process, as it helps to ensure that the audit is conducted in an effective and efficient manner. It also helps to ensure that the audit is conducted in accordance with the standards and guidelines set out by the relevant regulatory bodies

2.3 Audit methodology

Session duration: 170 minutes

Session specific objective: At the end of this session, participants will be able to develop audit project proposal using audit planner.

Activity 1: Question and Answer

List the purpose of clinical audit proforma

Instruction to conduct Gallery Walk

- Ask the questions to the whole group
- Encourage active participation
- Allow students to reflect/answer on the purpose of clinical audit
- Validate and summarize the answers provided

Time: 20min

Clinical audit proforma/planner is a document that outlines the criteria and processes for conducting a clinical audit. It is used to ensure that the audit is conducted in a consistent and systematic manner, and that all relevant information is collected and analyzed. The template typically includes sections for the purpose of the audit, the scope of the audit, the methodology used, the data sources, the data analysis, the results, and the recommendations.

The purpose of the audit should be clearly stated in the template. This should include the objectives of the audit, the expected outcomes, and the timeline for completion. The scope of the audit should also be outlined, including the population or sample size, the type of data to be collected, and the methods used to collect the data. The methodology should include the steps taken to ensure that the data is collected and analyzed in a consistent and reliable manner.

The data sources should be identified in the template, including the type of data, the sources of the data, and the methods used to collect the data. The data analysis should include the methods used to analyze the data, the results of the analysis, and any conclusions that can be drawn from the analysis. The results should be presented in a clear and concise manner, and should include any recommendations for improvement.

The template should also include a section for the recommendations. This should include any changes that need to be made to the process or procedures, any additional resources that may be needed, and any other actions that need to be taken to improve the quality of care. The template should also include a section for the implementation of the recommendations, including the timeline for implementation and any resources that may be needed.

It is important that data collected in the course of any clinical audit is precise and pertinent to the audit being performed. To ensure that data is collected appropriately, there are a number of details which need to be established at the outset. These are:

- The user group to be included, with inclusion/exclusion criteria defined.
- The consent required to access user group information.
- The healthcare professionals involved in the service user's care.
- The time period over which the criteria apply

Before data collecting begins, a structured strategy should be used to identify the pertinent data and make sure that the process is quick, accurate, and efficient. We should consider the sort of data we need (quantitative, qualitative, retrospective, or prospective), the data source, the design of the data collection tool, the sample size, the data gathering procedures and methods (manually, electronically, and for how long).

Selecting a sample or a population

It is initially required to define the population of concern before choosing a population or sample (target population). A population can be described as all the service users, occasions, cases, circumstances, or things that the audit is concentrating on. A sample is a subset of the target population.

The following should be taken into account when choosing which is appropriate for the goals of the clinical audit:

- The size of the population.
- Resources available for the audit
- Whether sampling or population is most appropriate?
- What sample selection method should be employed?

Sample selection methods

For audit reasons, it is frequently not practicable or necessary to collect data on every service user, event, or object; as a result, sampling is frequently needed.

Any sample that is chosen must accurately reflect the population being studied.

There are many other sampling techniques that can be used, however convenience sampling and random sampling are often employed techniques.

The most basic type of random sampling is picking service customers at random from a list of the entire population, such as every third, sixth, etc. case. This function is provided by the Hospital Inpatient Enquiry System (HIPE) (HSE, 2008). It is also possible to generate random numbers.

Convenience sampling is used as a quick and efficient method of conducting a sample survey.

It entails selecting the most accessible and nearby individuals to serve as responders; as a result, it does not yield results that may be considered to be representative, such as the first 10 cases to present after a certain period of time

Interval sampling – is often determined by a time period. For example, all cases in a specific timeframe

Sample size

Clinical audit is not a Research. Sample sizes for audit is frequently a compromise between the statistical validity of the results and practical difficulties around data collection, such as time, access to data, and expenses. It is about evaluating compliance with standards rather than providing new knowledge

For the purposes of the audit, a minimal amount of data may be adequate for many audit issues; however, if a disputed issue is being examined, a higher sample size may be necessary.

The sample size must to be adequate to produce significant results. The sample should allow for case mix adjustment when necessary. When possible, pre-existing data sets should be used in the clinical audit. The sample should be both large enough to be representative and small enough to allow for quick data gat hering. The sample will be time-driven in certain audits and numerical in others. One to three months should be sufficient for the majority of time-based audits.

Activity: Group work and presentation

- Being in your previous group formed for topic selection, and for the topic you have selected in your groups develop complete clinical audit proforma (proposal) composed of all the components of clinical audit using audit planner (Annex 1).
- The assignment will be given a day before to start in your groups
- After completion of the audit proforma present for whole class
- Reflect and discuss
- Wrap up with facilitators

Time: 120 minutes

Summary

The clinical audit proforma template is an important tool for ensuring that the audit is conducted in a consistent and systematic manner. It should be used to ensure that all relevant information is collected and analyzed, and that the results are presented in a clear and concise manner. The template should also include sections for the implementation of the recommendations, and for any additional resources that may be needed.

Chapter 3: Measure level of performance and making improvements

Duration: 10 ½ hours

Chapter description

This chapter is intended to emphasize on measure current practices of service delivery against the agreed standard, making change and sustain improvements. It also focuses on how to determine the target, ensuring changes made to be maintained for the service delivery. The changes made may be sustained by different approaches.

Primary objective: At the end of this session, participants will be able to make and maintain improvement in healthcare delivery by conducting clinical audit.

Enabling objectives:

At the end of this chapter, participants will be able to:

- Analyze types of data and data collection
- Interpret audit data and compare with agreed standards
- Draw conclusion from results of clinical audit data
- Plan for change implementation
- Apply results of audit for improving quality of service
- Conduct re-audit

Chapter outline

3.1 Data collection

3.2 Analysis of data and drawing conclusion

3.3 Making changes

3.4 Sustainability of changes

3.1 Data collection

Session duration: 150 minutes

Session specific objectives: At the end of this session, participants will be able to:

- Analyze data of clinical audit in a way that it can be applied for improving service delivery.

- Compare the data obtained from clinical audit against the agreed standard

Activity 1: Question and Answer

- Think of your awareness of clinical audit, and reflect how you will collect data, analyze and present data obtained from clinical audit.

  Time: 5 minutes

It is important that data collected in the course of any clinical audit is precise and pertinent to the audit being performed. To ensure that data is collected appropriately, there are a number of details which need to be established at the outset. These are:

- The user group to be included, with inclusion/exclusion criteria defined.
- The consent required to access user group information.
- The healthcare professionals involved in the service user's care.
- The time period over which the criteria apply

Planning data collection

Before data collecting begins, a structured strategy should be used to identify the pertinent data and make sure that the process is quick, accurate, and efficient

We should consider the sort of data we need (quantitative, qualitative, retrospective, or prospective), the data source, the design of the data collection tool, the sample size, the data gathering procedures and methods (manually, electronically, and for how long).

Data type

The audit question and objectives will determine the type of data that is needed. Data must be collected in a way that makes it possible to compare current practice to audit Standard because that is the purpose of data gathering

Data items

All information gathered must be pertinent to the audit's goals and objectives. Each data point must be sufficient and not excessive for measuring practice against the pertinent audit standards. This is equally significant. Data collection that is not necessary for measuring purposes is time-consuming, yields little to no utility, and may not be in line with information governance laws and procedures.

Definitions and instructions for data collection should be compiled once standards of best practice, audit criteria, estimated compliance rates, and known exceptions have been determined. For data collecting purposes, this entails identifying terms in the audit criteria and known exceptions, as well as specifying how and from where evidence should be gathered

Sources of data

The audit team should identify and agree upon the source of data for the audit. The cited source ought to deliver the most precise and comprehensive information as quickly as feasible. For the objectives of the clinical audit, appropriate and regularly obtained raw data from already existing sources should be used whenever possible as this prevents information and work from being duplicated and enables recurring data collection and reaudits with the least amount of effort.

Data collection strategy

The audit topic and objectives should be taken into account before choosing a data gathering strategy. The following factors play a role in choosing a data collection strategy for a clinical audit:

- Implementation timeline and resource availability.
- Which method of data collecting is most likely to produce accurate and comprehensive information?

Retrospective data collection

After a service user's therapy or care is complete, retrospective data is collected.

The fact that the data is already available and can be swiftly gathered is one benefit.

It may be possible to identify all service users who match the inclusion criteria, but this depends on the abili ty to identify pertinent service users using coding or other record-keeping systems.

However, the information may not be complete and reliable, and it can soon become outdated.

Prospective data collection

While treatment or care is being given, concurrent data is being collected. Planning for the recording of information on the care that will be provided is part of prospective audit. Sometimes, the words concurrent

and prospective are used synonymously. An audit should be conducted either prospectively or concurrently if the necessary data is not regularly gathered.

Time is needed to gather the data, and there is a chance that it will be biased if clinical practice is altered by the awareness that an audit is occurring when data is collected concurrently with patient treatment.

Activity 2: Group work

Be in your previous group formed for audit topic selection and:

- Determine the sample required for audit topic you have selected as priority based on its objective to improve service delivery.
- Reflect to the larger group and discuss how you determine the sample size for your audit project.

Time: 30 minutes

Who collects the data?

Depending on the audit, different people may be in charge of completing various data sets or more than one person may be used to gather data. There shouldn't be any ambiguity in terms. Additionally, everyone participating in data collecting should be aware of and understand whose responsibil ity it is for each piece, such as what information should be gathered and how.

Ensuring data quality

When data accomplishes its intended purpose, it can be deemed to be of high quality.

To ensure that data collectors have a solid knowledge of what, how, and when data needs to be gathered, each data item that has to be collected needs to have explicit descriptions. To reduce the likelihood of repor ting and input errors, frequent data quality checks should also be performed.

Summary

Data collection is an important part of any research project. It is important to select the most appropriate data collection method for the research question being asked. The data collected should be accurate, reliable, and valid. It is also important to consider the ethical implications of data collection.

3.2 Data Analysis and drawing conclusion

Session duration: 180 minutes

Session specific objective: At the end of this session, participants will be able to analyze and conclude data obtained by conducting clinical audit.

Activity 1: think pair share

- Think and discuss about data analysis in clinical audit.
- Think in your own for 2 minutes about analysis techniques for clinical audit and share in pair with participant next to you.
- Reflect your pair ideas in larger group and discuss.

Time: 10 minutes

A collectedfacts (data) must be transformed into useable information in order to determine the degree of co nformity with the established standard. The main aim of data analysis is to answer the questions posed by the audit objectives; highlighting areas of good practice and areas that require particular attention or improvement. The kind of information gathered determines the sort of data analysis, Simple averages and percentages to complex statistical methods can all be used in this. Complex statistical analysis is neither required nor appropriate for majority of audits. To encourage change, a straight forward, understandable analysis that is accessible to all parties involved in the delivery of care is needed. To carrying out an audit with a small dataset, a pen, paper and calculator may be all that is required to carry out a simple analysis. Alternatively, a spreadsheet program such as Microsoft Excel may be a useful tool to perform the calculations required and to produce tables, charts and graphs for presenting and comparing information in a simple, easy to understand format

Calculating compliance with clinical audit criteria

Identification of whether or not performance levels have been attained is the fundamental necessity of an audit. Calculating the proportion of cases that have adhere to each criterion is necessary for this. It is essential to determine both the total number of applicable cases for a criterion (the denominator) and th e total number of individuals within the denominator group that met the criterion in order to calculate the per centage (the numerator) the percentage is then calculated by dividing the numerator by the denominator and multiplying the answer by 100.

$$\frac{\text{Numerator}}{\text{Denominator}} \times 100$$

The denominator is the total number of cases to which the audit criterion applies. This figure is obtained by subtracting cases meeting any agreed exceptions for a particular criterion from the total number of cases which meet each of the inclusion criteria and none of the exclusion criteria.

Displaying data

The data should be presented in the simplest, clearest, and most efficient form possible to make it easier to derive conclusions from the analysis of the data. There are numerous ways to present data, like comparing data from one area to data from another, comparing results to an expected level of performance, or comparing the results of an audit today to those from earlier audits. The final step in the process (when relevant) is to determine why the standard was not fulfilled after results have been produced and the data have been analyzed against the standard. The audit team should thoroughly evaluate all findings in order to determine why compliance with clinical audit criteria wasn't achieved. To identify any examples that might still constitute appropriate care, specific cases where the provided care does not meet the requirements should be examined. Cases of unacceptable care should then be reviewed in order for the team to:

- Clearly identify and agree on areas for improvement identified by the clinical audit.
- Analyze the areas for improvement to identify what underlying, contributory or deep-rooted factors are involved.

Presentation of audit results

Any presentation of results should seek to maximize the audience's response to the clinical audit in order to generate conversation and encourage and support action planning**.** There are various different methods for the presentation of clinical audit results including:

- ❖ Visual presentations
- ❖ Written reports for submission to the relevant clinical lead, directorate or governance committee
- ❖ Verbal presentations at relevant meetings

Activity 2: Case study

- Be in your previous group and discuss the following case study.

case:

- There was a clinical audit conducted on 5 key national institute of health clinical excellence (NICE) guidelines relevant to health visiting and school of nursing practice. The aim of the audit was to ensure that health visitors and school nurses in the city and county localities have a working knowledge of 5 key NICE guidelines relevant to practice. (*See full clinical audit result in annex V*)

Based on the case (result of clinical audit), discuss in your group and:

- Analyze the clinical audit topic with topic selection criteria
- Evaluate their objectives
- Evaluate their data analysis and presentation
- Draw a conclusion from the case and plan for change implementation.
- Present your group result in class and demonstrate reflective feed back
- Wind up the case discussion with the facilitator.

Time: 100 minutes

Summary:

Overall, data analysis in clinical audit is an important tool for healthcare organizations to ensure that they are providing the best possible care to their patients. It can help healthcare organizations identify areas of improvement in clinical practice, identify areas of risk and potential areas of improvement, and identify areas of cost savings. By analyzing the data, healthcare organizations can develop strategies to improve the quality of care provided to patients and reduce costs. Drawing audit conclusions is an important part of the audit process. It is the process of summarizing the results of the audit and making a determination of the existing service delivery status. The final conclusion of the audit should be recommended to appropriate stalk holders for making changes and improve service delivery.

3.3 Making changes and improvements

Session duration:150 minutes

Session specific objectives: At the end of this session, participants will be able to apply the results of clinical audit to improve clinical service delivery and making changes on the area.

Activity 1: Write-pair-share

- Based on the awareness about clinical audit that, write down how can we use the results of clinical audit to bring change on clinical service delivery? Take 5 minutes to write down your ideas individually and re-write the ideas in peer with trainee next to you for another 5 minutes.
- Reflect your peer ideas and discuss in larger group.

Time: 15 minutes

"The most important part of the audit cycle is making change" -*Baker et al (1999).* "A clinical audit which does not result in improved services and increased patient safety is a waste of resources." -*Healthcare Quality Improvement Partnership (2014).*

Ashmore, Ruthven and Hazelwood (2011c) identify clinical audit as a change process, stating: 'audit that simply measures but does not drive change to address problems identified, is not good audit. All good audit projects must include a program of change activity and post-identification of the findings from audit, to ensure necessary changes happen.'

It is confirmed that clinical practice can be changed by clinical audit in order to enhance over all patient care and outcome. But change is not always simple. It is the most challenging.

To make it easy, implementing change after conducting a clinical audit typically involves the following steps:

1. Review audit results: Analyze the findings of the audit to identify areas for improvement.

2. Develop a plan: Create a comprehensive plan that outlines the changes to be made, who will be responsible for implementing them, and a timeline for completion.

3. Communicate the plan: Share the plan with relevant stakeholders, including staff, patients, and management, to ensure everyone is aware of the changes and understands their role in the process.

4. Implement changes: Put the plan into action and make the necessary changes to improve clinical practice.

5. Monitor and evaluate: Regularly monitor the impact of the changes and evaluate their effectiveness. This can be done through follow-up audits or other means of assessment.

6. Sustain changes: Once the changes have been implemented, work to ensure they become a permanent part of the clinical practice. This may involve updating policies and procedures, providing ongoing training and support, and making any necessary modifications to the plan.

When we create a change implementation project, we have considered the following in order to increase chance of success.

- Assure team members are inspired to enhance procedures. You are less likely to effect change if no one else is interested in the audit or if you are conducting it merely out of obligation.
- Include all of the important actors. If everyone who will ultimately make decisions alters
- All audit reports should be shared with the service, specifically with the relevant Head of Department, Head of Service and Governance Group. Where audits are instigated by a particular committee/group, reports should be submitted to the committee/group for review and action.

Clinical audit results may show areas of excellent or 'notable practice' and this should be acknowledged. For such audits there should be an explicit statement saying 'no further action required' in the audit summary report and a rationale why re-audit is not required. Clinical audit results may also identify 'areas for improvement' where the required standards are not being met.

Making recommendations for change

Activity 2: Group activity

be in group of 3-5 participants and discuss the following in your groups.

- How can we develop recommendations for change from clinical audit project.
- Techniques of assessing the root cause analysis (RCA) to analyze the exact pitfall of the healthcare care service detected by the clinical audit.
- Reflect your ideas and discuss in larger group

Time: 10 minutes

Change is often the most difficult part of the audit. When the audit team have developed the recommendations, decisions should be made on how changes can be introduced and monitored. Results should be used in conjunction with feedback and local consensus to change clinical practice and to improve standards. Priorities for action should be identified and these should be clearly documented. All audits should be accompanied by a quality improvement plan in order to achieve the required improvements in practice.

Before you make any changes, it's important to consider if you truly comprehend the underlying causes of shortcomings of the service identified by the audit. It could be possible to investigate this further using root cause analysis (RCA) and related methodologies.

5 whys?

It's really simple to use this method, which entails asking "why" frequently. To find a problem's underlying cause, repeat a process five times. For instance:

• The patient's skin cancer diagnosis came much later than expected. Why?

• The surgeon did not see the excision biopsy report. Why?

• The report was entered into the patient's records without the surgeon seeing it. Why?

• The filing was the receptionist's responsibility. Why?

● The junior doctors were busy with other tasks. Why?

The root cause - that the doctors' other tasks were seen as more important than filing. The system has now been changed. A copy of all biopsy reports is now sent to the consultant surgeon responsible for the patient and no reports are filed unless they have been signed by a doctor.

FISHBONE DIAGRAMS

Also known as ‘Cause and Effect’ diagrams, that shows breaking-down all the contributory causes of an observed effect, including less-obvious factors. This can help to decide which factors are most likely to be causing the problem and decide what you can address most effectively.

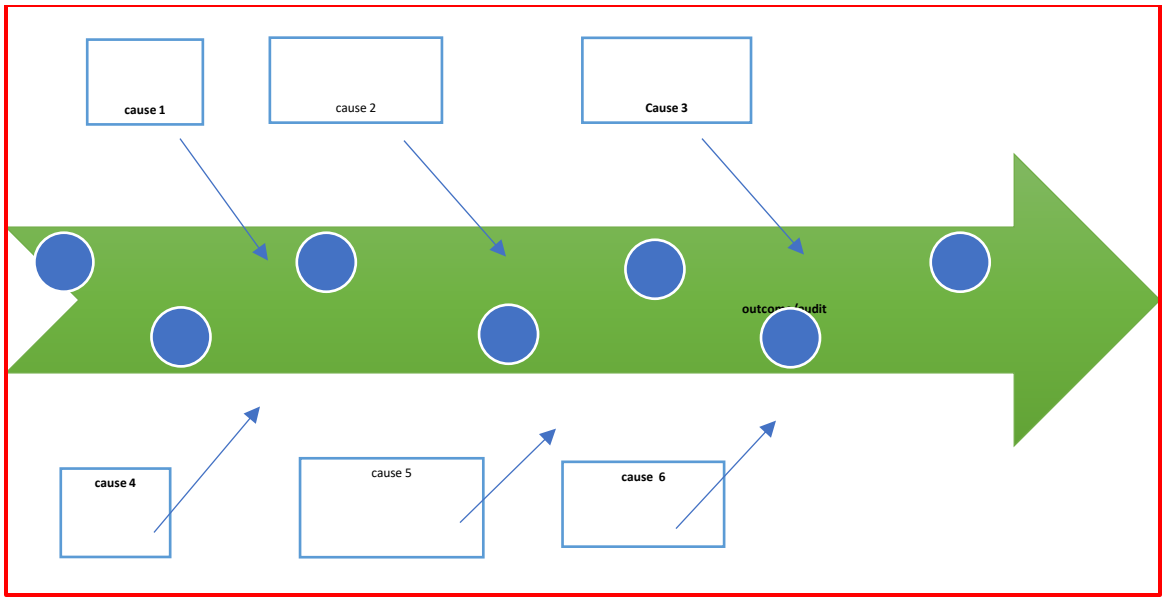

Figure 4: Fish-bone diagram

Activity 3: Group activity

Be in your previous group formed and.

- Describe each factor for undesirable outcome for clinical service area identified using FISH BONE DIAGRAM.
- Reflect your group idea and discuss in larger group.

Time: 15 minutes

PROCESS MAPPING

Examining patient journeys through the service provided in much more detail may be beneficial depending on the audit's complexity. For instance, if your audit shows that patients aren't getting their blood test results in a timely manner, you might be able to utilize process mapping to pinpoint the areas that are problematic and determine whether any processes can be skipped or rearranged to make the pathway more effective. The more you involve the staff people who will ultimately be accepting proposed changes at this stage, the more probable it is that all these exercises will involve greater collaboration with multidisciplinary teams.

Change process

Change process has three main stages.

1. Initiation - The process leading up to the change.

2. Implementation – The first experiences of change.

3. Continuation – The changes become embedded.

1. **Initiating change**

It needs to analyze the situation before you think implementing changes.

- Do people aware of the need for change? The presentation of your audit results should be used to notify people of the need for change and to 'sell' to them your recommendations for change. Willingness to change varies from person to person. For example, someone who has been working in a particular clinical area for a short period of time might be more open to the idea of change than someone who has been working there for a longer period.
- You may need to sell your proposal. An important factor to bear in mind is that, whilst clinicians will be interested in what the proposed changes might mean for their patients, they will probably be most concerned about the implications for them personally.
- People respond to different stimuli when it comes to thinking about change. For some, a shared vision of the future will suffice. Others will want to be persuaded by facts and figures.
- There may be individuals who will only change practice if a reward or penalty is at stake. Use power or influence where you can.
- It is important to have an opinion leader on your side especially when the barrier for change is cultural reason, because the majority of any group will accept changes in response to the action of opinion leaders, i.e. people who are well respected.

Useful tools for initiating change:

Before implementing change, there should be early plan and there are number of useful tools for his. The most common three popular tools which help to anticipate different reactions and encounter potential resistance are:

1. **TROPICS-** This is a good way to get a feel for the nature of a particular change and plan an appropriate strategy:

   T Time scales - Defined? Short or long term?

R Resources - What will be needed?

O Objectives - Are these quantifiable?

P Perceptions - Does everyone see this issue the same way?

I Interest - Who has an 'interest' in making change happen/keeping things the same?

C Control - Who holds the power?

S Source – Who is driving this proposal, internal or external source?

2. **Stakeholder Analysis** - This is a framework for thinking about where your colleagues might stand in relation to the proposed changes and the most appropriate approach for you to take with them.

Table 4: Stalk holder analysis framework

		Trust	
		High	Low
Agreement	High	**Allies** Ask for their advice and support Keep them informed	**Bedfellows** Make agreements, but keep an eye on them
	Low	**Opponents** Engage and negotiate	Enemies Isolate, out-maneuver or forget

3. **Forcefield Analysis** - This is a way of visually mapping out the forces that are likely to affect (help or hinder). You can develop a strategy to reduce impact of hindering/opposing factors and strengthen the supporting/driving forces by carefully identifying prons and cons. As a rule of thumb, it is better to reduce restraining forces, which can be rational or emotional, rather than increase driving forces. Driving and restraining forces might relate to an individual, staff group or to the organization as a whole.

   They can include the following questions:

   - Have past experiences of similar situations/changes affected people's views?
   - Is management supportive?
   - How unacceptable/undesirable is the current situation perceived to be?
   - Are there fears about increased workload?
   - Is there pressure for change from patients?

- Are there national policy requirements we must comply with?

Forcefield Analysis can be used effectively in combination with TROPICS and/or Stakeholder Analysis

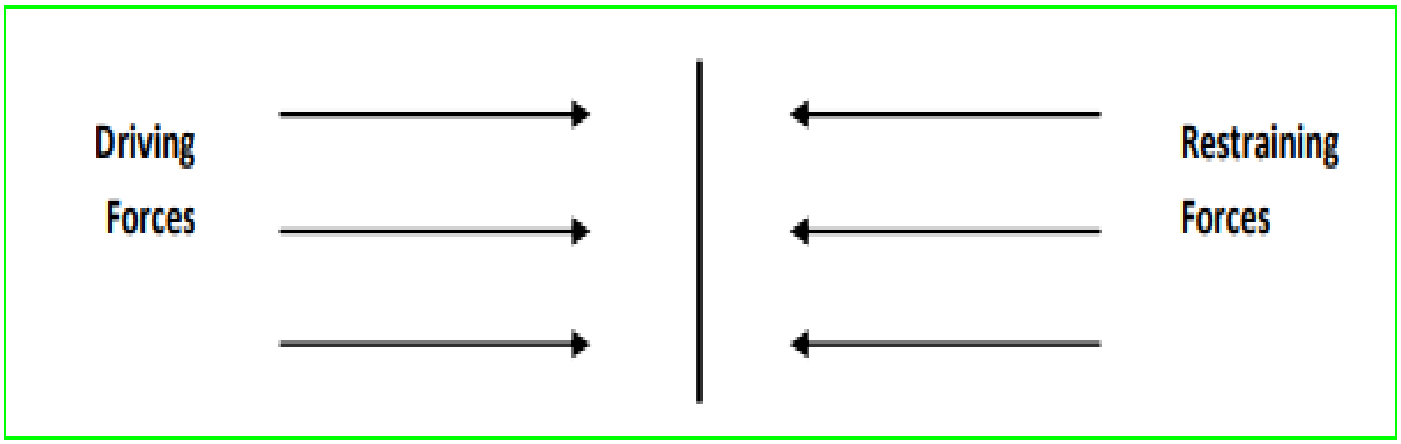

Figure 5: Forcefield analysis diagram

2. Implementing change

Assuming you have won your colleagues over to the idea of the proposed change, you will now want to implement it. You may need to plan the implementation phase, even if it is simply a question of purchasing a piece of equipment. You may need to break down the changes into manageable tasks and achievable targets. Crucially, it also means communicating, e.g. informing staff about what is going on and consulting them for their own ideas. Different objectives require different methods of communication. Sending out a newsletter about a new clinical guideline is, on its own, unlikely to change clinical practice. It is therefore important to consider whether or not there is a need for training and development, e.g. organizing briefings/ workshops. A useful rule to remember is that a strong action makes it easier to do the right thing and harder to do the wrong thing.

WEAK ACTIONS	STRONG ACTIONS
Raise staff awareness	Remove barriers to doing the work effectively
Remind staff	Redesign the work
Provide training	Supervise, monitor and feed back
Write a new policy	Use IT or technological solutions

3. Continuation of change

Even if you are successful in implementing changes, there is no guarantee that they will remain in place. People sometimes revert to their former working habits. Once again, communication is crucial, both with the

members of staff who are implementing the change and with management. Provide staff with evidence that the changes have had a positive impact through a re-audit. If other staff members are slow to come on board with the changes, can management encourage them to move their position?

REASONS WHY CHANGE SOMETIMES FAILS

Change can fail at any one of the stages of change process; frequent reasons for failure include a lack of resources, a lack of motivation, inadequate management of the process, or poor communication. Usually factors that hinder change can be addressed by proper planning of clinical audit project from the beginning to the implementation of audit findings.

- Create a multi-professional/multi-disciplinary audit team with a representative from each staff group involved in the care being audited; this will increase ownership of the problems and improve motivation for change.
- Ensure that people with authority to agree changes have involved.
- If you are likely to need resources to implement changes, ensure that management is in agreement with the aims of the project and will provide funds if necessary.

Summary:

Clinical audits are a systematic process used to evaluate and improve the quality of patient care by comparing actual practices to established standards and best practices. By conducting clinical audits, healthcare organizations can identify areas for improvement, make changes to their practices, and monitor the impact of these changes to ensure continued quality improvement.

3.4 Sustainability of changes

Session duration: 150 minutes

Session specific objectives: At the end of this session, participants will be able to **sustain improvement in clinical audit through effective planning, implementation, and continuous monitoring and evaluation of audit activities.**

Activity 1: think-pair-share

- Think in your own about the performance indicators used to evaluate the quality of care provided in our health institutions? Take 2 minutes to think your ideas individually and share in peer with trainee next to you for another 3 minutes.
- Reflect your peer ideas and discuss in larger group.

Time: 8 minutes

Performance Indicators

A performance indicator in clinical audit refers to a measure used to assess the quality of care provided by a healthcare organization or provider

Performance indicators can be used to track advancements brought about by quality improvement efforts. To track the execution of the improvement plans, a limited set of key performance indicators may be created for each quality improvement program. When undertaking quick re-audits, this technique can also be utilized to assess how changes have affected practice.

Some common performance indicators in clinical audit include:

1. Patient satisfaction scores

2. Clinical outcomes (e.g. mortality rates, readmission rates)

3. Process measures (e.g. compliance with evidence-based guidelines)

4. Resource utilization (e.g. length of hospital stay, cost per patient)

5. Patient safety measures (e.g. adverse event rates)

6. Patient experience (e.g. wait times, communication with staff)

7. Provider performance (e.g. adherence to protocols, documentation accuracy)

Clinical audit is an important process for evaluating and improving the quality of healthcare delivery. To evaluate the quality of a clinical audit, several factors should be considered such as the clarity and completeness of the audit's objectives, the validity of the data collected and analyzed, the reliability and accuracy of the audit findings, the effectiveness of the audit process in identifying and addressing areas for improvement, and the impact of the audit recommendations on patient outcomes. Additionally, the involvement of stakeholders, such as clinicians and patients, in the audit process can provide valuable

feedback and help to ensure that the audit is relevant and meaningful. Overall, a well-designed and executed clinical audit should demonstrate a positive impact on patient care and lead to continuous quality improvement in healthcare delivery.

To evaluate the quality of a clinical audit, the following factors can be considered:

1. Objectives: Are the objectives of the audit clearly defined and aligned with the overall goals of the healthcare organization?
2. Methodology: Is the methodology used for the audit evidence-based and reliable?
3. Data Collection: Is the data collected during the audit accurate, complete and relevant?
4. Analysis: Is the data analyzed in a systematic and appropriate manner to determine areas for improvement?
5. Recommendations: Are the recommendations made based on the findings of the audit evidence-based and practical?
6. Implementation: Is there a plan in place for implementing the recommendations and monitoring their impact?
7. Feedback: Is feedback provided to stakeholders about the results of the audit and the impact of the recommendations?
8. Continuous Improvement: Does the audit process include a mechanism for continuous improvement, such as regular review and updating of the audit methodology.

Activity 2: Role play

Perform a role-play where one participant will play a role of clinical auditor who conducted clinical audit on certain service area, one participant will play a role of reviewer of the audit conducted and the rest of the class will play the role of audit committee. The auditor will present findings of clinical audit while the reviewer and the audit committee will provide feed-back on quality of clinical audit.

Time: 20 minutes

Disseminate and celebrate success

Improvements in practice typically result from the completion of an audit cycle. It should be made known to all interested parties.

A successful audit in one service may be transferable to other parts of the service. Completed audits should be shared locally via the most appropriate mechanisms, including department quality and safety meetings, journal club meetings, the intranet, newsletters and local conferences and seminars. Consideration should also be given to sharing clinical audit work regionally and nationally through relevant journals, conferences and other media.

Re-audit

Another crucial step in the audit cycle is the re-audit, which needs to be done with specific interval commonly 3-12 months after the adjustments are made. Re-audit entails gathering a new set of data to assess the results of the changes after they have been put into place and determine whether more development is required. The audited statistics need to be comparable to those from the initial data collection phase. The audit phase completes audit cycle and link back to the first step of the cycle. Problem or objective identification

Re-audit involves collected a second state of data and analyze the result to determine whether the change has been made or not.

Audit is a continuous cycle. If a program of change activity has been undertaken after an initial audit reveals that desire performance levels are not being met, the audit should be repeated to see whether the changes made have improved care or whether additional changes are needed. This cycle is repeated until the desired performance levels are being achieved.

Summary:

Sustain improvement in clinical audit is the ongoing process of maintaining and improving the quality of healthcare services through ongoing evaluation and monitoring of clinical practices. This involves regularly reviewing and analyzing patient care processes to identify areas for improvement, implementing changes to

enhance patient outcomes, and continuously monitoring the effectiveness of these changes to ensure they are sustained over time.

This includes.

1. Developing and implementing a systematic approach to clinical audit.
2. Establishing a system of feedback and follow-up to ensure changes are implemented.
3. Establishing a system of continuous monitoring and evaluation.
4. Developing and implementing a system of accreditation and quality assurance.
5. Developing and implementing a system of data collection and analysis.
6. Developing and implementing a system of benchmarking and performance improvement to collaborate with other departments and stakeholders.
7. Developing and implementing a system of communication and education to provide training and support to staff
8. Establishing a system of rewards and recognition for successful clinical audit initiatives.

References

1. A Practical Guide To Clinical Audit
2. Clinical Audit by health quality improvement partnership (HQIP)
3. Principles for Best Practice in Clinical Audit – NICE
4. The Basics in Research Methodology: The Clinical Audit
5. Healthcare Audit Criteria and Guidance
6. Accreditation Standards for Hospitals (5th Edition)
7. Introduction to Evidence-Based Medicine
8. Evidence-Based Medicine – A Paradigm ReadyTo Be Challenged?
9. The Sample Size Calculator is available via:

   http://www.uhbristol.nhs.uk/forclinicians/clinicalaudit/how-to-guides/

Annexes

Annex I: Clinical Audit proforma

Clinical Audit Planning & Summary Form	Federal Ministry of Health Health Service Quality Directorate

Project Title: (Identified Audited practice to Improve surgical access – at least three topics)	Audit Ref No

Project Team

1. Senior Clinician	2. Project leader (junior person doing the data collection)
3. Name of person from the audited department	4. Others, as necessary

Background

What?

Why?

What benefits for the patient?

Aims& Objectives

Standards	Target	Evidence	Data source

Clinical Audit Planning & Summary Form	Federal Ministry of Health Health Service Quality Directorate

Project Title: (Identified Audited practice to Improve surgical access – at least three topics)	Audit Ref No

Audit Methodology

Audit Population

Audit Sample

Data collection method

Deadlines

Start date

Completion date

Presentation date

Main Results (compared to specific standards)

Clinical Audit Planning & Summary Form	Federal Ministry of Health Health Service Quality Directorate

Project Title: (Identified Audited practice to Improve surgical access – at least three topics)	Audit Ref No

Conclusions (highlight areas identified for improvement)

Areas of good practice

Areas which need to be improved

Action to be implemented	By whom	By When

Clinical Audit Planning & Summary Form		Federal Ministry of Health Health Service Quality Directorate
Project Title: (Identified Audited practice to Improve surgical access – at least three topics)		**Audit Ref No**
Re-audit required:- Yes / No		
Presentation **Date**		

Annex II: Gallery Walk

A gallery walk is a great way to engage participants in a training session. It can be used to introduce a new topic, review material, or even to assess understanding. Here are some instructions for conducting a successful gallery walk:

1. Prepare the space: Before the gallery walk, make sure the space is set up for success. Arrange the room so that participants can easily move around and view the materials. If possible, provide tables or stands to display the materials.

2. Prepare the materials: Gather the materials that you want to use for the gallery walk. These can include posters, handouts, pictures, videos, or any other visual aids. Make sure the materials are clearly labeled and organized in a way that makes sense.

3. Introduce the activity: Explain to the participants what a gallery walk is and how it will be conducted. Let them know that they will be moving around the room to view the materials and that they should take notes or ask questions as they go.

4. Divide into groups: Divide the participants into small groups and assign each group to a specific area of the room. Make sure each group has enough time to view the materials and discuss them.

5. Allow time for discussion: After each group has had a chance to view the materials, allow time for discussion. Encourage participants to ask questions and share their thoughts and ideas.

6. Wrap up: At the end of the gallery walk, review the materials and discuss any key points or takeaways. Make sure to thank the participants for their participation and encourage them to ask any remaining questions.

By following these instructions, you can ensure that your gallery walk is a successful and engaging activity for your training session. It can be a great way to introduce a new topic, review material, or assess understanding

Annex III: Tips for successful development of clinical audit

Top tips for the successful development of clinical audit data collection tools (*Developed by the Irish Clinical Audit Network 2006)*

Validity As in research, the data collection tool should have a high level of 'validity' i.e. it collects the data that it purports to collect. This is why it is important to start with clear clinical audit objectives and use evidence-based standards from the outset.

Reliability As in research, the data collection tool should have a high level of reliability, i.e. different personnel collecting the data will record the exact same or highly similar data using the tool. Inter-rate reliability: If data is being collected by many personnel it is important that a check is made to ensure that inter-rater reliability is high.

Sequencing Data collection questions should flow logically based on the sequence of events in the data source i.e. patient records etc. If feedback is sought from service users or staff members questions should be worded clearly and follow the same basic principles used in developing research interview schedules.

Quantitative versus qualitative clinical audit data collection Clinical audits can use either quantitative or qualitative data to meet the clinical audit objectives.

- The majority of clinical audits focus on quantitative data. Data collection tools of this type should ensure that clear closed questions are used i.e. Yes/No, Documented/Not documented etc.

- Where qualitative data is required it is necessary to develop data collection questions for use in interviews/focus groups or via questionnaire.

Drafting closed questions When drafting closed questions:

- use mutually exclusive options, for example Yes/No; Male/Female

- include an option for all responses, for example Yes/No/Don't know/Not applicable

- where appropriate, give more than two options, for example, 4 or 6 (an even number is better as this may assist selection of the middle option as a default). Reliability tends to rise as the number of options rise.

Presentation Ensure that the data collection tool is present in such a way that it is visually appealing to those collecting the data:

- Use standard fonts.

- Ensure that the font size is readable.

● Do not overcrowd each page – ensure that there is space for recording all relevant information and additional comments.

Comments Ensure that there is sufficient space for including comments during data collection Include the date of data collection Ensure that the date of data collection is recorded clearly on each data collection sheet. This is particularly helpful when data is collected over an extended timeframe.

Annex IV: Think-pair-share

Name _____ Date_____

Think-Pair-Share

Read the following question or problem:

Think

On your own, write three ideas you have about this question or problem:
1._____
2._____
3._____

Pair

Discuss your ideas with a partner. Put a check by any ideas, above, that your partner also wrote down. Then, write down ideas your partner had that you did not have:
1._____
2._____
3._____

Share

Review all of your ideas and circle the one you think is most important. One of you will share this idea with the whole group. As you listen to the ideas of the whole group, write down three more ideas you liked:
1._____
2._____
3._____

Annex V: sample clinical audit for case study

Annex III: sample clinical audit for case study

An audit of the knowledge of 5 key National Institute of Health and Clinical Excellence (NICE) clinical guidelines relevant to Health Visiting and School Nursing practice- 2012/13 (436)

Audit period: September 2012-November 2012

Report date: February 2013

Executive Summary

Why the audit was undertaken

The overall aim of this audit was to ensure that health visitors and school nurses in FYPC have a working knowledge of 5 key NICE guidelines relevant to practice that have been published between 2007 and 2010. In order to achieve this it was necessary to establish the level of knowledge in 3 areas for each of the conditions:

Key signs and symptoms

When further action is needed

The correct management and advice to give.

A similar audit had been undertaken in the previous county organisation which highlighted concerning gaps in knowledge, although this was considered to be a new baseline audit as it used a different tool and methodology, and was carried out over a new organisation.

How the audit was carried out

A questionnaire was designed with questions around each of the above areas of knowledge, asking for one response per question. Staff were asked to complete the questionnaire at professional meetings, where attendance was expected to be high. They had no prior warning of the audit.

Key findings

A number of areas of concern were identified, with compliance being rated as minimal in accordance with LPT clinical audit policy (dated April 2012) Some of these need addressing as they indicate practice which is contrary to that recommended in the NICE guidance as well as in the Standard Operating Procedures for health visitors. Full analysis of the results was made difficult by the high number of multiple responses. It also became clear that for the school nurses, compliance was below that originally expected for some conditions, but as these related to situations they would not meet on a regular or frequent basis it could be

argued that these should not have been included. However, a surprisingly low number of staff said they were aware of where to access NICE guidelines, despite all having access to the internet.

Key actions

The key recommendations arising from the results of the audit of the health visitors are around education and raising awareness of correct diagnosis and management of jaundice and of eczema. Some actions, such as training and circulation of new operating procedures, have started to take place and will go some way towards addressing the knowledge gaps in these areas. Further learning points will be disseminated by line managers to individual teams and discussed at reference group meetings.

Re-audit date

The audit will be repeated between September and November 2013.

Aim

The overall aim of the audit is to ensure that health visitors and school nurses in the city and county localities have a working knowledge of 5 key NICE guidelines relevant to practice.

Objectives

The objectives are:

to establish whether health visitors and school nurses are aware of the key signs and symptoms of the condition specified in each guideline.

to establish whether they can identify when further action is needed.

to establish whether they are aware of the correct management and advice to give.

Criteria & standards

The following NICE guidance were audited:

National Institute for Health and Clinical Excellence

CG 47 Feverish Illness in children under 5 (May 2007)

CG57 Atopic Eczema in Children (December 2007)

CG84 Diarrhoea and vomiting in children under 5 (April 2009)

CG98 Neonatal jaundice (May 2010)

CG99 Constipation in children and young people (May 2010)

None of the above is in the process of being updated, although CG47 is scheduled for review in May 2013.

Table 5 - Criteria and standards

Criteria	Standards

Health visitors and school nurses can correctly identify what NICE stands for, and where to access the clinical guidelines. Questions 1 and 17. All participants in the audit are registered nurses and should be aware of current guidelines and where to access them as part of their professional code of conduct.	The expected standard is 100%.
Health visitors and school nurses can make a correct diagnosis, identify when further action is needed and identify what the correct management advice is for the following 4 conditions: diarrhoea and vomiting in children under 5, feverish illness in children under 5, atopic eczema and constipation. Questions 2,3,4and 8-16,	The expected standard is 95%.

Although this is less than the 100% compliance expected for each of the guidelines, this will allow for practitioners who do not use this knowledge on a daily basis and may need to seek further advice themselves before offering advice to clients. It will also allow for practitioners who are new in post and who will not have received emails raising awareness of NICE guidelines over recent months.

Health visitors can correctly identify key clinical signs of prolonged neonatal jaundice, when further action is needed and what the correct management is. Questions 5,6 and 7,	The expected standard is 100%.

As the NICE guideline audit tool sets compliance at 100% then the standard of knowledge must also be 100%. This is an area of practice which health visitors could encounter on a daily basis while visiting children at home, so a sound working knowledge is essential.

The following rating was applied to the results to highlight any particular areas of concern, which is consistent with the trust agreed approach and the recommendation of the Clinical Audit Support Centre.

Full compliance	Partial compliance	Minimal compliance
$x \geq 95$ (or 100)%	70%$\leq x <$94(or 99)%	$x <$ 69%

Method

Questionnaires were completed by staff during professional locality meetings from September to November 2012. The audit was listed as an item on the agenda for team meetings as NICE but there was no pre-warning about the audit. Staff were given time to complete the questionnaire at the beginning of the meeting, ensuring a high response rate, and handed them straight back to the manager who sent them to the audit team for analysis. Locality and team meetings are seen as a priority for all health visiting and school nursing staff so a high representative sample should have been achievable. Completed audit forms were returned to the audit office and were inputted and analysed using Excel.

Sample & data source

A minimum target of 160 would have given a confidence level of 95% with a 5% error margin. The total population is approximately 270.

Audit type

This is a new baseline audit.

See **Error! Reference source not found.** on p.**Error! Bookmark not defined.** for a copy of the audit tool.

Service areas / teams included

The audit was carried out on teams from health visiting and school nursing across all localities in the city and the county. It had been aimed at qualified members of the team (i.e. nurses registered with the NMC) but some were completed by assistants within the teams and by staff with nursery nurse qualification. It was decided to exclude these results from the final analysis as the expected compliance for this group of staff would be different, due to the differing roles and responsibilities. Registered nurses working within health visiting teams were included.

Although the questionnaire asked for locality to be identified this proved to be problematic when analysing the results due to differing terminology used. For example there are two North West and two South teams, one each in the county and the city. It was decided therefore to analyse results without taking locality or neighbourhood into account as the expected compliance was the same across the whole area

Findings

Responses received

A total of 147 questionnaires were received from qualified staff, 111 from health visiting teams and 36 from school nurse teams. This was lower than expected but could be accounted for by anecdotal reports of team managers not being informed about the audit, lack of mutually convenient times for the audit team to attend the team meetings, and some teams declining to participate. Two questionnaires were returned with all answers blank, and were therefore excluded, as were those which did not indicate which professional group they belonged to. Some questionnaires were completed by community nursery nurses but these were not included in the analysis as the audit was looking at the knowledge of NMC registered nurses. This response rate still gives a confidence level of 95% with a margin of error of 7.15%.

Audit findings

Although it had originally been planned for the questionnaires to be electronically analysed on FORMIC this proved to be problematic due to the high numbers of staff ticking more than one box for the answer. Each question then had to be looked at to decide clinically whether to count the answer as right or wrong, depending which boxes were ticked. For example, in question 7, if someone ticked A or B as well as the correct answer, C, then they would be counted as correct, because the advice in A and B could also form

part of the overall management, but would not be correct on their own. Similarly, if someone ticked an answer which was definitely incorrect as well as the correct one then they would be marked wrongly

Table 2 Audit results for health visiting teams (in comparison with thresholds of compliance)

Criteria	Expected standard	Number answered correctly	Compliance
Question 1: NICE stands for National Institute for Health and Clinical Excellence	100%	62	56%
Question 2: Gastroenteritis alone should be suspected if there is a sudden change to loose or watery stools.	95%	106	95%
Question 3: Which if the following children with gastroenteritis can be safely managed at home? An 18 month old child who has had diarrhoea for 6 days.	95%	93	84%
Question 4: Which of the following is the correct advice to a mother of a 1 year old with gastroenteritis, no risk factors and no signs of dehydration? Encourage breastfeed/other milk feeds and fluid intake, advice about Red Flag symptoms and how to get immediate help if they develop.	95%	99	89%
Question 5: Which of the following is NOT a sign of prolonged jaundice? Yellow colouration of blanched skin in a term baby aged 10 days.	100%	84	76%
Question 6: Which of the following will raise your concern about a baby with jaundice? Dark urine which stains the nappy.	100%	107	96%
Question 7: If a baby has prolonged jaundice at the primary visit which of the following described the correct management? Contact children's day ward at the LRI to make a referral, arrange the appointment and inform the parent(s).	100%	89	80%
Question 8: Which of the following can be used to diagnose a feverish illness? Parental report of a fever.	95%	15	14%
Question 9: Which of the following is a Red Flag symptom in a child with feverish illness (which would cause you to refer to a paediatric specialist urgently)? Weak or high pitched cry.	95%	104	94%

Criteria	Expected standard	Number answered correctly	Compliance
Question 10: Which of the following is a correct piece of advice for parents of a child with a fever (with no red or amber symptoms)? Look for signs of dehydration, non-blanching rash, encourage more fluids than usual and to seek further advice if they feel the child is getting worse.	95%	99	89%
Question 11: A diagnosis of atopic eczema can be made when a child has an itchy skin condition and which of the following? History of dry skin in the last 12 months, visible flexural dermatitis involving skin creases and allergic rhinitis.	95%	94	84%
Question 12: A referral to specialist dermatological advice is necessary if: Atopic eczema is associated with severe or recurrent infections.	95%	106	95%
Question 13: Which of the following describes the correct management for a 2 year old with dry skin, and occasional appearance of red patches of itchy skin behind the knees? Emollients, mild topical corticosteroids only where and when the skin is red and itchy, and continue with emollients even when fully resolved.	95%	85	77%
Question 14: Which of the following children is constipated? A 3 year old who passes 2 hard large stools per week, straining to do so and who has had constipation in the past.	95%	110	99%
Question 15: When assessing a child, which of the following findings is a Red Flag indicating an underlying disorder rather than idiopathic constipation? Ribbon-shaped stool and abdominal distension with vomiting.	95%	103	93%
Question 16: Which of the following describes the most appropriate dietary and lifestyle advice for parents of a 5 year old child with idiopathic constipation (in addition to management with laxatives)? Fluid intake of at least 1200ml as drinks daily, recommend foods such as fruit and vegetables, high-fibre bread and wholegrain cereals, daily physical activity and negotiated follow-up and support.	95%	105	95%
Question 17: Are you aware of where to access the NICE guidelines quick reference guides? Yes	100%	94	85%

Table 3 Audit results for school nursing teams (in comparison with thresholds of compliance)

Criteria	Expected standard	Number answered correctly	Compliance
Question 1: NICE stands for National Institute for Health and Clinical Excellence	100%	24	67%
Question 2: Gastroenteritis alone should be suspected if there is a sudden change to loose or watery stools.	95%	29	81%
Question 3: Which if the following children with gastroenteritis can be safely managed at home? An 18 month old child who has had diarrhoea for 6 days.	95%	25	70%
Question 4: Which of the following is the correct advice to a mother of a 1 year old with gastroenteritis, no risk factors and no signs of dehydration? Encourage breastfeed/other milk feeds and fluid intake, advice about Red Flag symptoms and how to get immediate help if they develop.	95%	24	67%
Question 8: Which of the following can be used to diagnose a feverish illness? Parental report of a fever.	95%	2	6%
Question 9: Which of the following is a Red Flag symptom in a child with feverish illness (which would cause you to refer to a paediatric specialist urgently)? Weak or high pitched cry.	95%	22	61%
Question 10: Which of the following is a correct piece of advice for parents of a child with a fever (with no red or amber symptoms)? Look for signs of dehydration, non-blanching rash, encourage more fluids than usual and to seek further advice if they feel the child is getting worse.	95%	24	67%
Question 11: A diagnosis of atopic eczema can be made when a child has an itchy skin condition and which of the following? History of dry skin in the last 12 months, visible flexural dermatitis involving skin creases and allergic rhinitis.	95%	34	94%
Question 12: A referral to specialist dermatological advice is necessary if: Atopic eczema is associated with severe or recurrent infections.	95%	34	94%
Question 13: Which of the following describes the correct management for a 2 year old with dry skin, and occasional appearance of red patches of itchy skin behind the knees? Emollients, mild topical corticosteroids only where and when the skin is red and itchy, and continue with emollients even when fully resolved.	95%	23	64%

Criteria	Expected standard	Number answered correctly	Compliance
Question 14: Which of the following children is constipated? A 3 year old who passes 2 hard large stools per week, straining to do so and who has had constipation in the past.	95%	34	94%
Question 15: When assessing a child, which of the following findings is a Red Flag indicating an underlying disorder rather than idiopathic constipation? Ribbon-shaped stool and abdominal distension with vomiting.	95%	35	97%
Question 16: Which of the following describes the most appropriate dietary and lifestyle advice for parents of a 5 year old child with idiopathic constipation (in addition to management with laxatives)? Fluid intake of at least 1200ml as drinks daily, recommend foods such as fruit and vegetables, high-fibre bread and wholegrain cereals, daily physical activity and negotiated follow-up and support.	95%	36	100%
Question 17: Are you aware of where to access the NICE guidelines quick reference guides? Yes	100%	29	81%

How compliance was calculated

$$\text{Compliance} = \frac{\text{number of YES}}{\text{number of YES and NO and BLANK}} \times 100$$